Lec-5

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Number System

① Binary number System → base 2 [0 to 1]
⑪ decimal number System → 10 → [0 to 9]
⑪ octal number System → 8 [0 to 7]

$378 = 3 \times 10^2 + 7 \times 10^1 + 8 \times 10^0$

```
 1 1
 2 3 4
 5 7 8
-------
 8 1 2
```

12 % 10 = 2
12 / 10 = 1
11 % 10 = 1
11 / 10 = 1
8 % 10 = 8
8 / 10 = 0

```
   1 1 1
 1 1 0 1 1 1
 + 0 1 0 1 1
 -----------
 1 0 0 0 1 0
```

2 % 2 = 0
2 / 2 = 1
3 % 2 = 1
3 / 2 = 1

```
2 | 39   1
2 | 19   1
2 |  9   1
2 |  4   0
2 |  2   0
2 |  1   1
     0
```

100 111

78

```
2 | 78   0
2 | 39   1
2 | 19   1
2 |  9   1
2 |  4   0
2 |  2   0
2 |  1   1
     0
```

$378 = 3 \times 10^2 + 7 \times 10^1 + 8 \times 10^0$

```
2 | 39   1 x10^0
2 | 19   1 x10
2 |  9   1 x10^2
2 |  4   0 x10^3
2 |  2   0 x10^4
2 |  1   1 x10^5
     0
```

sum = 1x10^0 + 1x10 + 1x10^2 + 0 + 0 + 1x10^5

mul = 1 x10 x10 = 10^3 4 5 6

while(n > 0) {
rem = n % 2
sum = sum + rem x mul
n = n / 2
mul = mul x 10
}

```
100 000
    1 0 0
      1 0
        1
----------
100 111
```

```
public static void main(String[] args) {
		// TODO Auto-generated method stub
		int n = 67;
		int sum = 0;
		int mul = 1;
		while (n > 0) {
			int rem = n % 2;
			sum = sum + rem * mul;
			n = n / 2;
			mul = mul * 10;
		}
		System.out.println(sum);

	}
```

→ 10^2 3 4
10
10
2

sum = 1x10^0
= 1 + 10
= 11 + 100
= 111 + 100000
= 100 111

```
2 | 39   1 x10^0
2 | 19   1 x10^1
2 |  9   1 x10^2
2 |  4   0 x10^3
2 |  2   0 x10^4
2 |  1   1 x10^5
     0
```

```
10 | 100 111   1 x2^0
10 | 100 11    1 x2^1
10 | 1001      1 x2^2
10 | 100       0 x2^3
10 | 10        0 x2^4
10 | 1         1 x2^5
     0
```

```
public static void main(String[] args) {
		// TODO Auto-generated method stub
		int n = 10101;
		int sum = 0;
		int mul = 1;
		while (n > 0) {
			int rem = n % 10;
			sum = sum + rem * mul;
			n = n / 10;
			mul = mul * 2;
		}
		System.out.println(sum);
	}
```

src or des → 10

while(n > 0) {
rem = n % des ✓
→ sum = sum + rem x mul
n = n / des ✓
mul = mul * src
}

3 → 6
10

1 0 1 1 0 1 0 0
$2^7\ 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$

Data Type

primitive — non primitive

byte
Short
int
long
} integer

float
double } floating number
boolean → True/false
char →

int x = 5
1 KB = 1024 mb

byte → 8 bit → 1 byte
Short → 16 bit → 2 byte
int → 32 bit → 4 byte
long → 64 bit → 8 byte
Float → 32 bit → 4 byte
double → 64 bit → 8 byte
boolean → 1 bit
char → 16 bit → 2 byte

byte

_ _ _ _ _ _ _ _
2 2 2 2 2 2 2 2 $2^8 = 256$

1 1 1 1 1 1 1 1
0 0 0 0 0 0 0 0

[0, 255]

-127 0 127

[-127, 127]

256

Sign
-ve ← 1
+ve ← 0
MSB

1 1 1 1 1 1 1
1 1 1 1 1 1 1

① _ - - - - - - -

1 000 0000
-0 ← 0 000 0000
+0

① 1's compliment

0 1 0 1 0 1 0 — x
1 0 1 0 1 0 1

0 0 0 0 0
1 1 1 1 1

2's → compliment
only for neg

0 1 1 1 1 1 1 1
$2^7\ 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$

0 0 1 0 1 0 1 0 1 — x
1 1 0 1 0 1 0 1 0
→ 0 0 0 1
1 1 0 1 0 1 0 1 1 → x

[-128, 127]

-ve ← 1 0 0 0 0 0 0 0
+ve ← 0 1 1 1 1 1 1 1

1000 0000
0111 1111
0000 0001
1111 111
1 000 0000

byte → 8 bit → -128 to 127 → -2^7 to $2^7 - 1$
Short → 16 bit → -2^{15} to $2^{15} - 1$
int → 32 bit → -2^{31} to $2^{31} - 1$
long 64 bit → -2^{63} to $2^{63} - 1$

```
1 1 0 1 0 0 1 0 0
x
```

```
1 0 0 1 0 1 1 0 0
```

1 0 1 0 0 1 0 0
0 1 0 1 1 0 1 1
1 1 1
0 1 0 1 1 1 0 0

0 0 1 0 1 1 0 0 — 44

2's zero

byte b = 5
byte b = -5

0 0 0 0 0 1 0 1
1 1 1 1 1 0 1 0
1 1 1 1 1 0 1 1

2^{16}

256

26 → a to z
26 → A to Z
52

2^{16}

Unicode
Ansiee ✓

97 a 65 A 0 48
98 b 66 B 1 49
99 c 67 C 2 50
100 d 68 D 3
| | 4
z Z 9

```
public static void main(String[] args) {

		for (byte b = 0; b < 128; b++) {
			System.out.println(b);
		}
	}
```

-128 < 128

b = byte(b+1)
-128

0
1
2
3
126
127
-128
-127